LEARNER GUIDE

Where does the line cross the y-axis?

Straight Lines and Connected Representations. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

In the equation y = mx + c, the number c is the y-intercept: the value of y when x = 0 — the point where the line crosses the y-axis. (A straight line runs on for ever in both directions, so it has no real start or end; but this one crossing point is worth naming.) In this lesson the gradient is held fixed, so you can study c on its own. You slide c and watch the line move straight up and down without changing its steepness; you slide a line onto a target crossing point; you find c from a graph, an equation and a table; and you build a family of parallel lines that share a gradient but cross the y-axis in different places. The question you are exploring is: what does c control when nothing else has changed?

What you are learning to notice

- That c is the y-intercept: the value of y when x = 0, where the line crosses the y-axis.
- That changing c slides the line up or down but does not change its steepness.
- How to read c from a graph, from an equation, and from a table.
- That lines with the same gradient but different c are parallel.

Moving through it, one step at a time

In the Intercept Isolator, the gradient is locked. Drag c and watch the line slide while its steepness stays the same, and the crossing point (0, c) move up or down. In Slide to a Target, drag c until the line passes through the marked point. In Find the Intercept, read c from a graph, an equation and a table. Then build three parallel lines with the same gradient and different starts. There is no rush; use the coordinate grid on the Scratchpad to draw and label intercepts.

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (reading coordinates, and substituting x = 0 into a rule to find the y-intercept), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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